

The Integrating Factor Method to Solve Linear First Order Ordinary Differential Equations

Benjamin Abshire[†]

February 25, 2026

I Introduction

Let $y(x)$ be an unknown function such that

$$y'(x) = x \tag{1}$$

Equation 1 is a “differential equation” since it involves the unknown function $y(x)$ and its derivatives. It is “ordinary” since it involves regular derivatives of a function with only one independent variable, which in this case is x . Equation 1 is “first order ” since it involves the first derivative. All together, it is a “first order ordinary differential equation”.

II The Integrating Factor Method

Let $p(x)$ and $q(x)$ be given functions. Then, we can define a “linear” first order ODE,

$$y' + p(x)y = q(x) \tag{2}$$

Where y, y' are the unknown functions. The ODE is a linear equation since it is a linear combination of the unknown function, and the coefficients are functions of the independent variable, x . Formally,

$$a_n(x)y^n(x) + a_{n-1}(x)y^{n-1} + \dots + a_2(x)y''(x) + a_1(x)y'(x) + a_0(x)y(x) = a(x)$$

The Integrating Factor Method can be used to solve all linear first order ODEs of the form of equation 2. If we let

$$\mu = e^{\int p(x)dx}$$

Then $\mu' = \mu p(x)$ which when multiplying equation 2 by μ we get

$$\mu y' + \mu' y = \mu q(x)$$

Which allows us to find the solution by using the formula

$$\int (\mu y)' = \int \mu q(x)dx \implies \mu y = \int \mu q(x)dx \tag{3}$$

[†]Definitions and example problems from MATH 32044: Ordinary Differential Equations instructed by Dr. Todd Neal and *Lectures on Differential Equations* by Phillip Korman. Kent State University.

III Examples

Example 1. $y' + 6xy = 8x$

Solution. Let $p(x) = 6x$ and $q(x) = 8x$. Then, let

$$\mu = e^{\int 6x dx} = e^{3x^2}$$

So,

$$\int (\mu y)' = \int \mu q(x) dx \implies e^{3x^2} y = \int e^{3x^2} 8x dx \implies e^{3x^2} y = \frac{8}{6} e^{3x^2} + C$$

So solving for y

$$y = \frac{\frac{8}{6} e^{3x^2} + C}{e^{3x^2}}$$

Example 2. $y' + \frac{x}{x^2+1}y = \frac{x}{x^2+1}$

Solution. Let $p(x) = q(x) = \frac{x}{x^2+1}$. Then, let

$$\mu = e^{\int \frac{x}{x^2+1} dx} = e^{\frac{1}{2} \ln |x^2+1|} = e^{(\ln |x^2+1|)^{\frac{1}{2}}} = (x^2 + 1)^{\frac{1}{2}}$$

So,

$$\begin{aligned} \int (\mu y)' = \int \mu q(x) dx &\implies (x^2+1)^{\frac{1}{2}} y = \int (x^2+1)^{\frac{1}{2}} \cdot \frac{x}{x^2+1} dx \implies (x^2+1)^{\frac{1}{2}} y = \int \frac{x}{(x^2+1)^{\frac{1}{2}}} dx \\ &\implies (x^2+1)^{\frac{1}{2}} y = (x^2+1)^{\frac{1}{2}} + c \implies y = \frac{(x^2+1)^{\frac{1}{2}} + C}{(x^2+1)^{\frac{1}{2}}} \end{aligned}$$

Example 3. $y' + \frac{1}{x+2}y = 3x - 2$, $y(0) = 6$. Note that this is called an “Initial Value Problem” (IVP) since we are given a value of the unknown function at a point.

Solution. Let $p(x) = \frac{1}{x+2}$ and $q(x) = 3x - 2$. Then, let

$$\mu = e^{\int \frac{1}{x+2} dx} = e^{\ln |x+2|} = |x+2| = (x+2)$$

We can rid of the absolute value since at $x = 0$, $|x+2| = 2 > 0$. So,

$$\int (\mu y)' = \int \mu q(x) dx \implies (x+2)y = \int (x+2)(3x-2) dx \implies (x+2)y = x^3 + 2x^2 - 4x + C$$

Plugging in for $x, y = 0$

$$\implies (0+2)6 = 0 + C \implies C = 12$$

Giving us the final solution,

$$y = \frac{x^3 + 2x^2 - 4x + 12}{x+2}$$